

Caffeine & the Heart

A student guide to what caffeine does to your heart and blood pressure, and how to manage it.

PART 1

How Caffeine Affects Your Heart & Blood Pressure

Caffeine — particularly in large amounts, like those found in energy drinks and pre-workout powders — puts real stress on the heart and blood vessels. This section covers the three main ways that happens, plus a list of other effects worth knowing about.

01 — It Raises Your Blood Pressure

Caffeine causes the blood vessels in your body to tighten up.* It does this by blocking a natural chemical in the brain that usually keeps blood vessels open and relaxed. When those vessels tighten, your heart has to push harder to move blood through them, which drives your blood pressure up. In young people, one large energy drink can raise blood pressure noticeably within half an hour to an hour of drinking it.

02 — It Can Make Your Heart Beat Faster or Irregularly

Caffeine wakes up the nervous system, which tells the heart to pump more blood. A cup of coffee usually does not cause much of a change, but energy drinks contain much higher doses. At those levels, the heart can become overworked — leading to a racing heartbeat,* skipped or extra beats,* or an uneven rhythm.* This is more likely to happen during or after exercise, when the heart is already working hard.

03 — It Can Make Blood Vessels Less Flexible Over Time

The inside walls of your blood vessels are normally smooth and stretchy, allowing them to widen or narrow depending on what the body needs. High doses of caffeine can temporarily damage this ability,* meaning the vessels stay stiff even when they should be relaxing. Over time, if someone regularly drinks large amounts of caffeine, the vessels can become permanently stiffer — making the heart work harder every day, not just when caffeine is in the system.

OTHER EFFECTS TO BE AWARE OF

- **Fight-or-flight response** — Caffeine pushes the body into an alert, stressed state — the same feeling you get when you are frightened or anxious. The calming, ‘rest and digest’ side of the nervous system gets switched off.
 - **Stacking ingredients** — Most energy drinks do not just contain caffeine — they also include other stimulants like taurine and guarana.* These can make caffeine’s effects on the heart even stronger.
 - **Dehydration** — Caffeine makes you urinate more, which reduces the amount of fluid in your blood. With less blood to pump, the heart has to beat faster to keep everything running normally.
 - **Reduced blood flow to the brain** — Caffeine can slow down how quickly blood moves through the vessels in your head, which is one reason people sometimes feel dizzy or lightheaded after a strong energy drink.
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WHERE TO FIND RESEARCH PAPERS

RECOMMENDED DATABASES

PubMed / MEDLINE — *The most reliable database for medical and health research.*

Scopus / Web of Science — *Good for finding widely-cited science papers across many fields.*

Google Scholar — *A broad starting point — make sure to filter results by 'peer-reviewed'.*

SEARCH STRINGS

Copy and paste these directly into the search bar of any database above:

1. ("caffeine" OR "energy drinks") AND ("heart rate" OR "tachycardia" OR "blood pressure" OR "hypertension") AND ("young adults" OR "adolescents")
2. "caffeine" AND "cardiovascular system" AND "hemodynamics" AND "endothelial function"
3. "adenosine receptor antagonism" AND "caffeine" AND "blood pressure"

PART 2

How to Reduce the Negative Effects

There are practical steps — both in the moment and over the longer term — that can reduce the strain caffeine puts on the heart. They are grouped here by how quickly they are needed.

IF YOU HAVE ALREADY HAD TOO MUCH

- **Drink water** — Caffeine removes fluid from your body, so drinking water helps replace it. More fluid in the blood means less strain on the heart, and it also helps your body process and clear the caffeine faster.
- **Replace salts and minerals** — If you are sweating, shaking, or feeling nauseous after a large caffeine dose, drinking something with electrolytes* — like a sports drink or coconut water — can help the heart settle back into a normal rhythm.
- **Breathe slowly and deeply** — Taking slow, deliberate breaths switches on the calming part of your nervous system, which naturally brings the heart rate down. Even a few minutes of this can reduce the racing or anxious feeling caffeine causes.
- **Stop exercising** — If your heart is already stressed from caffeine, adding physical effort on top makes it worse. Sit or lie down and let the body recover.

LONGER-TERM HABITS

- **Spread it out, do not stack it** — Having a small amount of caffeine at a time — say 50 mg — is much gentler on the heart than drinking a 300 mg energy drink in one go. Spacing doses out gives your body time to adjust.
- **Check what you are drinking** — Many drinks contain more caffeine than people realise. Pre-workout powders and some sodas can contain hidden amounts, and guarana* — a common additive — is itself a source of caffeine that is often not included in the total shown on the label.
- **Do not mix caffeine with alcohol** — Alcohol makes you feel less alert, which can hide how much of an effect caffeine is having. This often leads people to drink far more of both than they intended, which is genuinely dangerous for the heart.
- **Cut down slowly if you are a heavy user** — If you drink a lot of caffeine every day and want to stop, reducing the amount gradually over one to two weeks is much easier on the body than stopping suddenly. Stopping suddenly often causes bad headaches and tiredness.

WHERE TO FIND RESEARCH PAPERS

RECOMMENDED DATABASES

CINAHL — *Focuses on nursing and health care — good for practical treatment and patient advice.*

Cochrane Library — *Publishes summaries of the best available research on health topics, including lifestyle changes.*

PubMed — *Covers the medical and scientific side of caffeine effects and management.*

SEARCH STRINGS

Copy and paste these directly into the search bar of any database above:

1. ("caffeine toxicity" OR "caffeine overdose" OR "energy drink adverse effects") AND ("management" OR "intervention" OR "treatment")
2. "lifestyle modification" AND "caffeine intake" AND "blood pressure reduction"
3. ("hydration" OR "water intake") AND "caffeine clearance" AND "pharmacokinetics"

Terms Used in This Guide

The following words appear with an asterisk () in the text above. Plain-language definitions are provided here.*

Blood vessels tightening (vasoconstriction) — When the tubes that carry blood around your body squeeze narrower. This makes the heart push harder, raising blood pressure.

Racing heartbeat (tachycardia) — When the heart beats faster than normal — usually above 100 beats per minute at rest. It can feel like your heart is pounding or fluttering.

Extra or skipped beats (extrasystoles) — Additional heartbeats that happen outside the heart's normal rhythm. They often feel like a brief 'thud' or a moment where the heart seems to pause.

Uneven rhythm (arrhythmia) — When the heart beats in an irregular pattern — too fast, too slow, or in an inconsistent way. Most are harmless, but some can be serious.

Blood vessel wall damage (endothelial dysfunction) — The inner lining of blood vessels loses its ability to relax and widen on demand. This makes the vessels stiffer and harder for blood to flow through smoothly.

Guarana — A plant found in South America whose seeds naturally contain caffeine. It is added to many energy drinks and supplements, but its caffeine content is often not included in the total milligrams listed on the label.

Electrolytes — Minerals dissolved in the blood — mainly sodium, potassium, and magnesium — that help control how muscles contract, including the heart muscle. Lost through sweat; replaceable through food, water, or sports drinks.

This guide is intended as a structured academic reference. Always consult peer-reviewed primary sources for university-level research.